

2001 Business Plan for
Performance-Based Software Development

(Performance-Based Contract Model)

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Document note

This English Edition is a content-oriented translation of the 2001 source document. The original Japanese diagrams, charts and page layouts are retained as visual references where reproducing every graphical label in English would alter the historical source. The Japanese original should be consulted for the original presentation and exact graphical details.

1. Overview of the Performance-Based Software Development Business

1-1. Business Description

Instead of charging for conventional development costs, revenue is generated as a **success fee** when the customer's profit, based on the original objective of the system investment, actually increases. The proportion of the fee varies according to the degree of risk shared with the customer. Revenue is intended to continue each year while the customer's profit remains above the pre-introduction level.

1-2. Business Background

Economic conditions have become increasingly uncertain because many complex factors operate on a regional, national and global scale. Companies are becoming less willing to make IT investments whose effects are unclear.

The document also points to growing requirements for safety and reliability, clearer operating requirements, reliable financial reporting, and explicit demands concerning the effects of system investment. These requirements extend from system planning and construction through maintenance, operation and post-implementation evaluation.

1-3. Need for the Business

Customer perspective: Customers can make investments proactively even without having sufficient funds initially; they can expect continuing IT investment effects; and they can direct funds generated by those effects toward other investments.

Development-side perspective: The aim is to create demand rather than merely respond to social demand, shifting from system construction toward business creation. The model seeks to move from revenue based on man-month accumulation to a revenue structure that is not proportional to headcount, and from order-volume dependence to continuing revenue based on customer profit, thereby strengthening the financial structure.

Industry perspective: The model is intended to strengthen organizations and human resources, raise the international competitiveness of software development, encourage an early transition away from labor-intensive work in anticipation of demographic change, and contribute to the competitiveness of Japanese industry.

1-4. Objective of the Revenue Structure

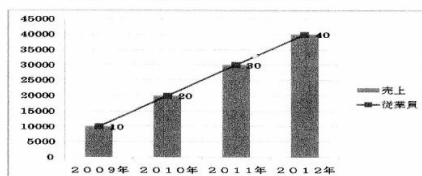
(1) Shift from a revenue structure based on man-month unit prices and accumulated man-months

In the conventional software-development business, revenue increases roughly in proportion to the number of employees, while sales per employee remain relatively low. In the performance-based model, revenue is linked to the customer's profit, so revenue is not proportional to employee headcount.

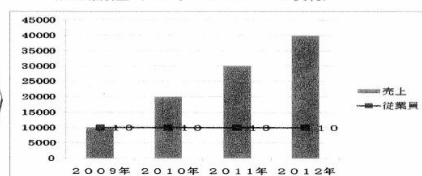
1-4. 成功報酬型ソフト開発事業における収益構造の狙い

(1) 人月単価及び人月積み上げによる売上構造からの転換

従業員の数に売上も比例し、一人当たりの売上げも低い。
人月計算（従来型ソフト開発ビジネス）



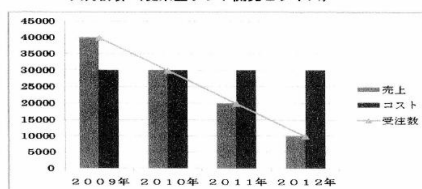
顧客の利益が関係しており、従業員数と売上げは比例しない。
成功報酬型（パフォーマンスベース契約）



(2) 受注減による資金ショートからの脱却（財務体質の強化）

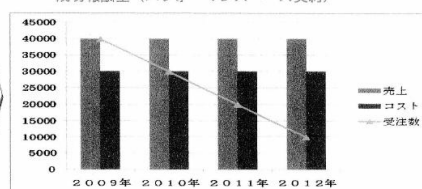
受注が減れば、コストは固定のため資金ショートとなる。

人月計算（従来型ソフト開発ビジネス）



顧客の利益が伸びれば売り上げは続く。

成功報酬型（パフォーマンスベース契約）



平野システム設計事務所 成功報酬型（パフォーマンスベース契約）ソフト開発事業プラン 無断転載・複写を禁ず

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Original charts: comparison of conventional man-month billing and the performance-based contract model.

(2) Escape from cash shortages caused by declining orders

When orders decline under a conventional model, costs remain relatively fixed and a cash shortage can result. Under the performance-based model, revenue can continue as the customer's profit grows.

2. Requirements for the Performance-Based Software Development Business

2-1. Concepts and Elements for Transformation

(1) From system construction to business construction — from manufacturing to value-added business

The software industry combines manufacturing and value-added business, making it inherently difficult. The system provider must plan and evaluate how the delivered system contributes to profit. This requires selecting an investment-evaluation method at the planning stage and clarifying the developer's own responsibility so that profit, quality and cost awareness become part of the organization.

(2) From man-month accumulation to an effect-based process

The conventional process has tended to focus on the steps required to build a system, using man-month accumulation as the basis. The proposed approach shifts toward a process based on investment effects: select technical means and architectures that can achieve the required effect with the minimum necessary scope, combine them appropriately, and establish continuous improvement so that construction can be based on measurable effects.

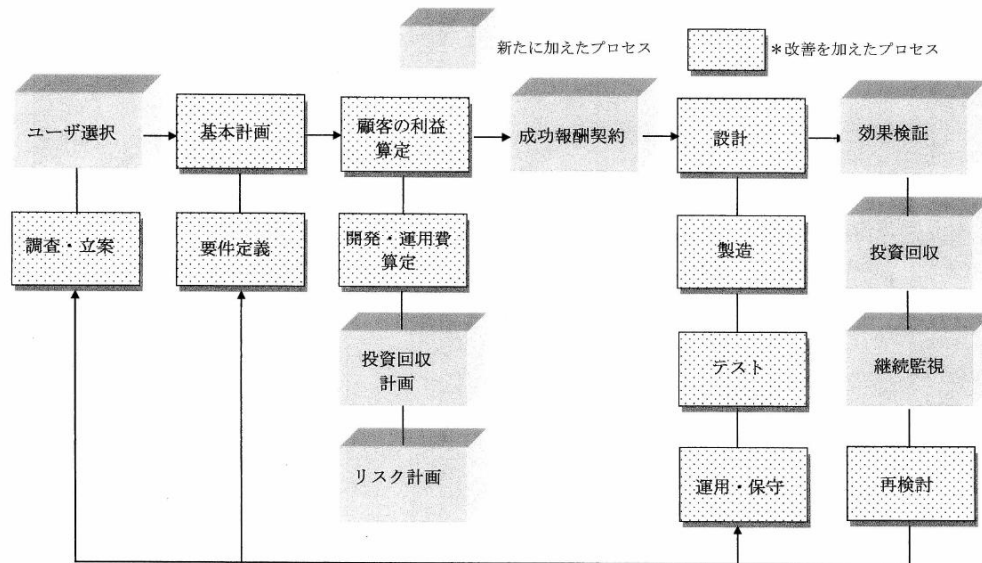
2-2. Business Process Overview

The business process is to be transformed from a process for constructing systems into a process for constructing the business itself.

The proposal retains the conventional system-contracting process while adding new processes needed to connect system development with business effects and performance-based revenue.

2-2. ビジネスプロセス概要

- ・ システムを構築するプロセスからビジネスを構築するプロセスへ転換する。
- ・ 従来のシステム受託プロセスに改善を加え、新たなプロセスを加える。



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Original process diagram retained as a visual reference.

2-3. Essential Objectives and Means of Realization

The document identifies the following objectives: high value, high quality, short development time, low cost, high productivity, and safety.

Software development depends on appropriately educated and trained people whose skills match the purpose of the project, including project managers, system planners and application engineers.

It also depends on practical organizational structures and appropriate processes, supported by management engineering, software engineering, quality engineering and safety engineering.

Management requires clear evaluation methods and quantitative methods for measuring results and promoting improvement. Tools should strengthen human resources, manage work, and systematize development work.

For low cost, short development time and high quality, reuse should be promoted through componentization and pattern-based approaches across planning, design, production and testing. The document also emphasizes understanding systems and maintenance through analysis, restructuring and reuse.

2-3. 欠かせない目的と実現手段

高 付 加 価 値
高 期 品 開 質
短 間 開 発
低 コ ス ト
高 生 産
安 全 性

手 段	必 要 性	内 容
人材	ソフトウェア開発は目的に合わせ教育・訓練された人材に依存する（成果は人が生む）	プロジェクトマネージャー システムプランナー アプリケーションエンジニア等
方法論	ソフトウェア開発は適切な組織構造に基づく実践的かつ適切な工程に依存する	経営工学 ソフトウェア工学 品質工学、安全工学等
管理	どのように人によって成果をあげることができるか及び改善を進める	評価方法 計量方法の明確化
ツール	人的資源の品質強化と作業管理および作業のシステム（ファクトリー）化	ビジネスプロセス作成、開発ツール、構成管理、テスト、プロジェクト管理、配布、運用管理、グループウェア等
再利用可能（部品化・パターン化）	低コスト、短期開発、高品質実現	工程別に部品化を進める 戦略立案、設計、製造、テスト
リエンジニアリング	システムの理解を高めること及び保守の自動化推進	解析、リストラクチャリング、リバーサスエンジニアリング、移行

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Original objectives-and-means table retained as a visual reference.

2-4. Departmental Planning Overview

Organization: Rather than organizing work strictly by industry-specific vertical divisions, teams are formed by professional roles and assembled in a task-force manner. Knowledge sharing is emphasized to improve work efficiency and speed.

Education and training: Results are treated as a habit. Knowledge and tools required for each role should become usable almost reflexively through training.

Human-resource system: Define the knowledge, skills and behavioral characteristics required for strong performance in each role; establish performance goals based on financial and non-financial criteria; align compensation with performance; and make evaluation criteria and processes transparent.

Human-resource acquisition: Consider speed, quality, demographic aging, declining population and diversity. The plan mentions acquisition of existing software companies, development of students, home-based work and acceptance of foreign personnel.

Facilities: Use locations with good mobility, introduce appropriate tools in offices, and establish environments that support development across multiple sites.

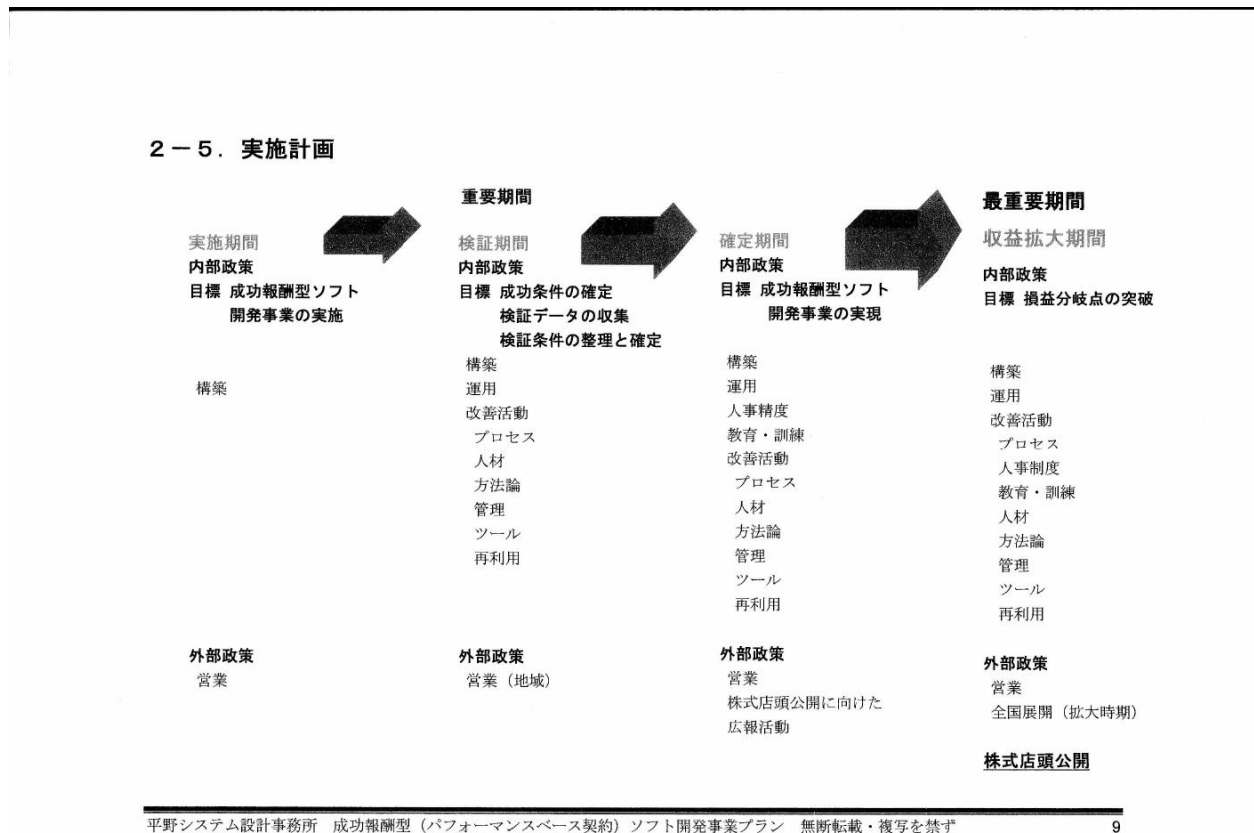
Financial planning: Consider the financial, customer, internal-process, and learning-and-growth perspectives, including use of the Balanced Scorecard.

Funding uses: Initially invest in people, education, hardware, operating systems, middleware, tools, networks, offices and reusable components in order to accelerate the business.

Risk planning: Identify and measure hazards and uncertainty, reduce uncertainty while following rules, and treat business opportunities as possibilities for higher returns while recognizing their associated risks.

2-5. Implementation Plan

The implementation plan is structured around successive periods: implementation, verification, confirmation, and revenue expansion. Internal policies cover business targets, success conditions, verification data, system construction and operation, improvement activities, personnel, processes, training, tools, methods and reuse. External policies include sales activities and broader expansion.



Original implementation-plan diagram retained as a visual reference.

2-6. Funding Policy

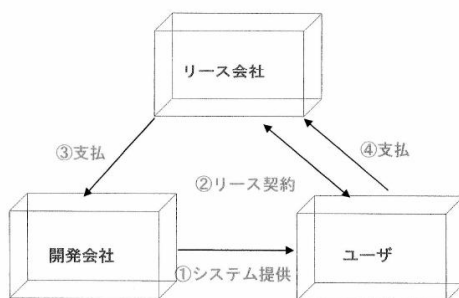
(Methods for dealing with cash shortages)

The proposal is to enter into a lease agreement with a leasing company in order to reduce near-term risk, prevent a slowdown in development caused by cash shortages, and improve cash flow. The diagram contrasts the existing payment flow with the proposed new payment flow involving the leasing company, development company, user and success-fee payments.

2-6. 資金政策

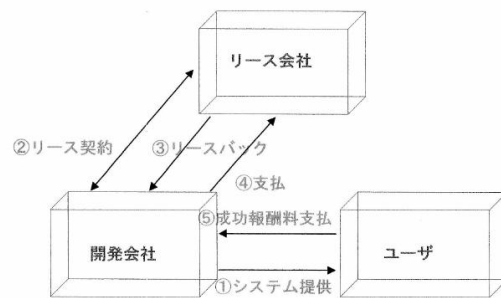
(資金ショート対処方法案)

(1) 今までの支払いの流れ



(2) 新たな支払いの流れ

開発元がリース会社とリース契約を結び、当面のリスクを軽減し、資金ショートによる開発スピード低下を防ぐとともに、キャッシュフローも高める財務政策をとる。



Original payment-flow diagrams retained as a visual reference.

2-7. Financial Plan

The financial statements in the source are illustrative rather than actual figures because revenue depends on how the user applies the system. When preparing a financial plan, it is important not only to determine when the investment will be fully recovered, but also to estimate the growth and continuity of revenue after recovery.

The plan also emphasizes reducing development and operating costs and considering the number of users that can realistically be acquired in accordance with the organization's capabilities.

The source provides multi-year illustrative sales, profit, funding and investment-recovery calculations. Because the OCR of the original table is imperfect, this English Edition does not reproduce the numerical table as an exact translated spreadsheet; the original Japanese table remains the authoritative version for the figures.

2-7. 財務計画

以下の財務諸表は、ユーザのどこにシステムを活用するかにより収益は異なるので実際の数字ではない。財務計画を立てるうえで投資回収完了の時期のみきわめも大事であるが、**回収後の収益の伸びと継続の見極めが重要なポイントである**。また開発・運用におけるコスト削減及び自社の組織力に応じた獲得ユーザ数も重要視しながら、財務計画に含めることも重要なポイントとなる。

(年度別売上・利益計画)

製品・サービス名	2013年度	2014年度	2015年度	2016年度
リースバック	250,000	356,250	450,000	637,500
ソフト開発成功報酬料		2013年 85,000	2013年 85,000	2013年 85,000
(プロジェクト別に損益分岐点の早期の突破と利益の伸びと継続が目標であるが、無理なプロジェクトは起こさない)			2014年 118,750	2014年 118,750
獲得ユーザ数 (総ユーザ数)	10社 (10社)	15社 (25社)	20社 (45社)	2015年 150,000 30社 (75社)
売上高計 (千円)	250,000	441,250	653,750	991,250
当期利益 (千円)	0	35,000	82,500	142,500

(資金計画)

主 な 内 容	2013年度	2014年度	2015年度	2016年度
システム開発 (人件費)	200,000	286,250	360,000	517,500
ツール・部品化・教育	50,000	70,000	90,000	120,000
リース支払 (5年)		2013年 50,000	2013年 50,000	2013年 50,000
コストを前年度より5%下げの方針で実施			2014年 71,250	2014年 71,250
合 計 (千円)	250,000	406,250	571,250	848,750
自己出資				
VC出資	250,000			
借入金				
自己資金				
合 計 (千円)	250,000	0	0	0

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Original financial-plan table retained as the authoritative numerical reference.

Presentation Purpose and Representative's Career

Presentation purposes: (1) funding and investment, including establishment preparation funds and initial operating funds; (2) business partnerships, including a data center and a major local wholesaler.

Representative: Naohiko Hirano

Born in 1957. Graduated from Kyushu Electronic Computer Technical School in 1979. Joined CSK (Computer Service) in 1979. From 1982 onward, worked independently in planning, design, production, testing, migration, operation and standardization.

The career record includes design of a securities online system, planning and management of a total system project for a major local wholesaler, planning and management of an outsourcing business, work on customer and sales-information sharing for a major railway company and its group, business and information-strategy planning for group companies, a group family-card project, and planning of business models using contactless IC cards, including financial business.

Part-time lecturer at Kyushu Electric Technical School (1994–2002), teaching system development. From 1992, research focused on realizing effectiveness and efficiency in information-system planning, development and maintenance, leading to the creation and presentation of the performance-based (performance-based contract) software-development business plan.

The original document contains personal contact information. It is intentionally not reproduced in this English Edition.

Supplement 1 — Eliminating Ambiguity in Planning

1. Clarify assumptions and premises used in calculations

When calculating the effects and costs of an information system, clearly define the premises and assumptions used for evaluation and calculation.

2. Clarify areas of indirect effects

When effects depend on people using the system, extend the analysis to the personnel system and quantify the results.

3. Fulfill appropriate operating requirements

To realize planned information-system effects, define operating requirements with the user department as the principal actor and align them with the customer's organizational capability. The document cautions against planning systems that exceed the organization's ability to operate them.

4. Clarify the degree of contribution

Analyze reductions in variable and fixed costs, increases in sales, new and repeat customers, entry into new business areas, and increases in profit based on the value chain. Align the analysis with management direction and establish priorities for using IT as a competitive advantage.

5. Clarify methods for verifying contribution

At the basic-planning stage, define how, when and by whom effects will be verified and which evaluation methods will be used. State these matters in the contract and verify them during operation according to the original plan. The source refers to IT investment-management frameworks and activity-based costing as examples.

6. Execute contracts based on consensus

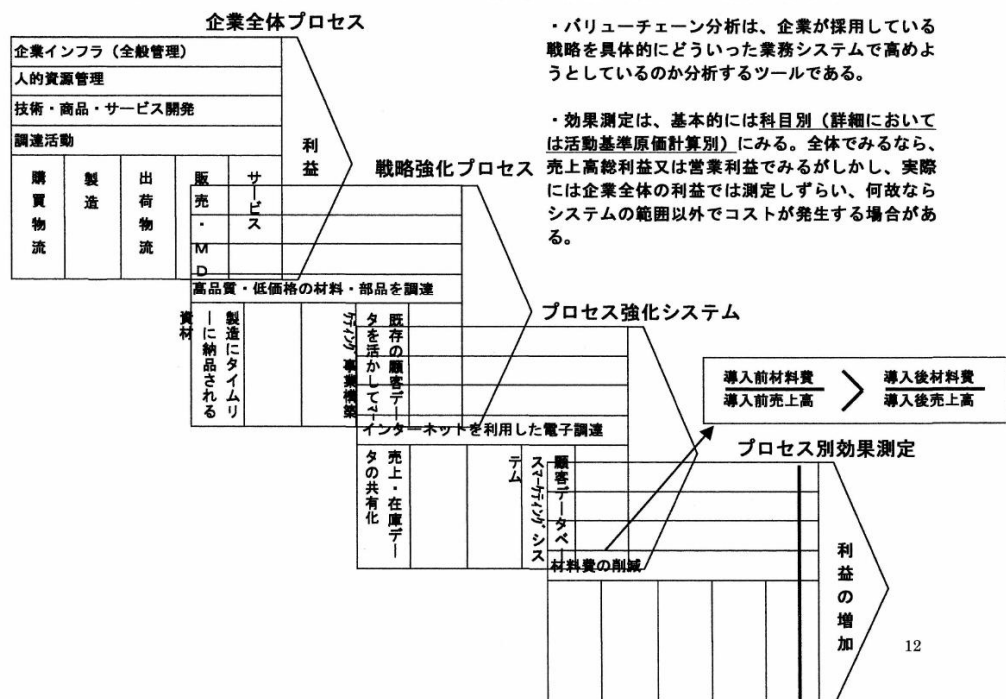
Obtain consensus from top management and the relevant user departments, specify what both the customer and developer must do, and execute a legally valid contract.

Supplement 2 — Example of Effect Verification Using a Corporate Value Chain

Corporate-process / value-chain analysis is presented as a tool for analyzing how the strategy adopted by a company is to be strengthened through specific business systems.

Effects are basically examined by account and, in greater detail, by activity-based costing. At the overall level, sales, gross profit or operating profit can be considered. The source cautions, however, that measuring the effect using total company profit alone can be difficult because costs may arise outside the scope of the system being evaluated.

補足資料1 企業のプロセス（バリューチェーン（価値連鎖））を用いた効果の検証例

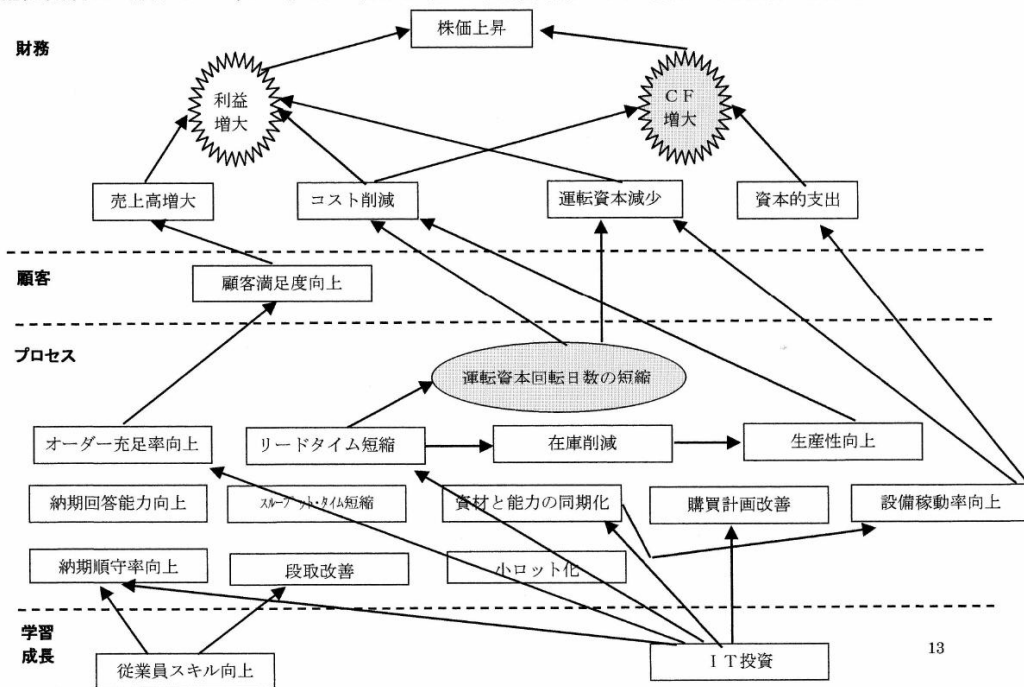


Original value-chain effect-verification example retained as a visual reference.

Supplement 3 — Example of Effect Verification Using a Cash-Flow / Balanced-Scorecard Causal Model

The model links financial outcomes with customer satisfaction, process improvements, and learning-related factors. Examples shown in the source include reductions in working-capital turnover days, improved order-fulfillment and delivery-response capability, improved purchasing planning, reduced inventory, improved equipment utilization, productivity improvement, and ultimately increased cash flow and profit.

補足資料2 キャッシュフロー（CF）とバランスコアード因果関係モデルを用いた効果の検証例

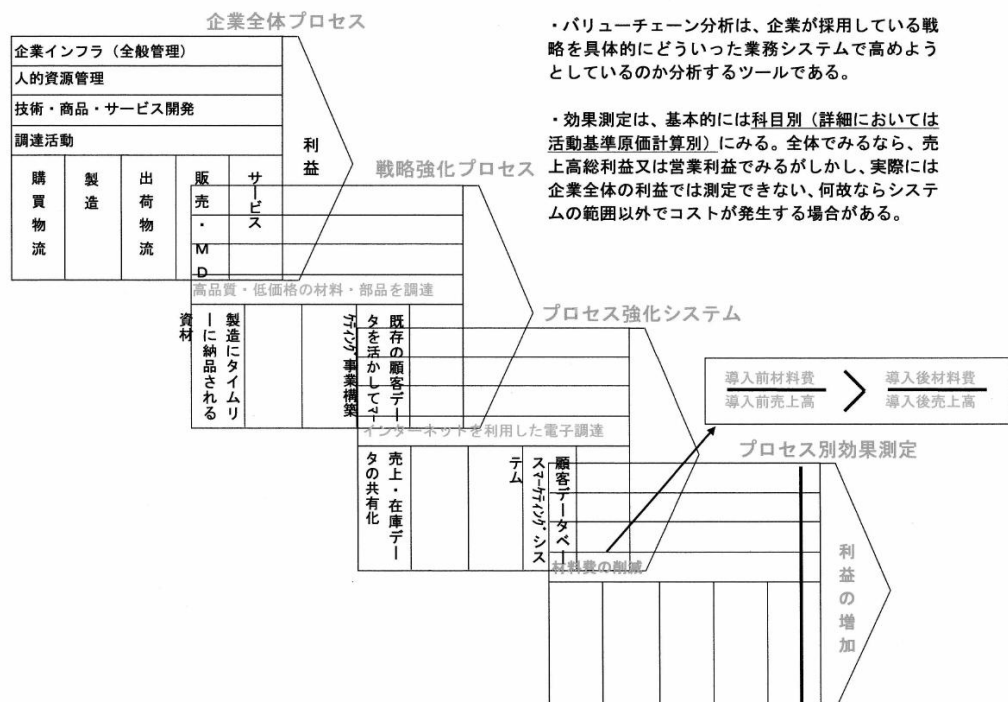


Original causal-model diagram retained as a visual reference.

Supplemental Effect-Verification Example

The source includes another corporate value-chain example. It emphasizes tracing effects through business processes rather than treating the whole-company profit figure as the only measurement.

補足資料3 企業のプロセス（バリューチェーン（価値連鎖））を用いた効果の検証例



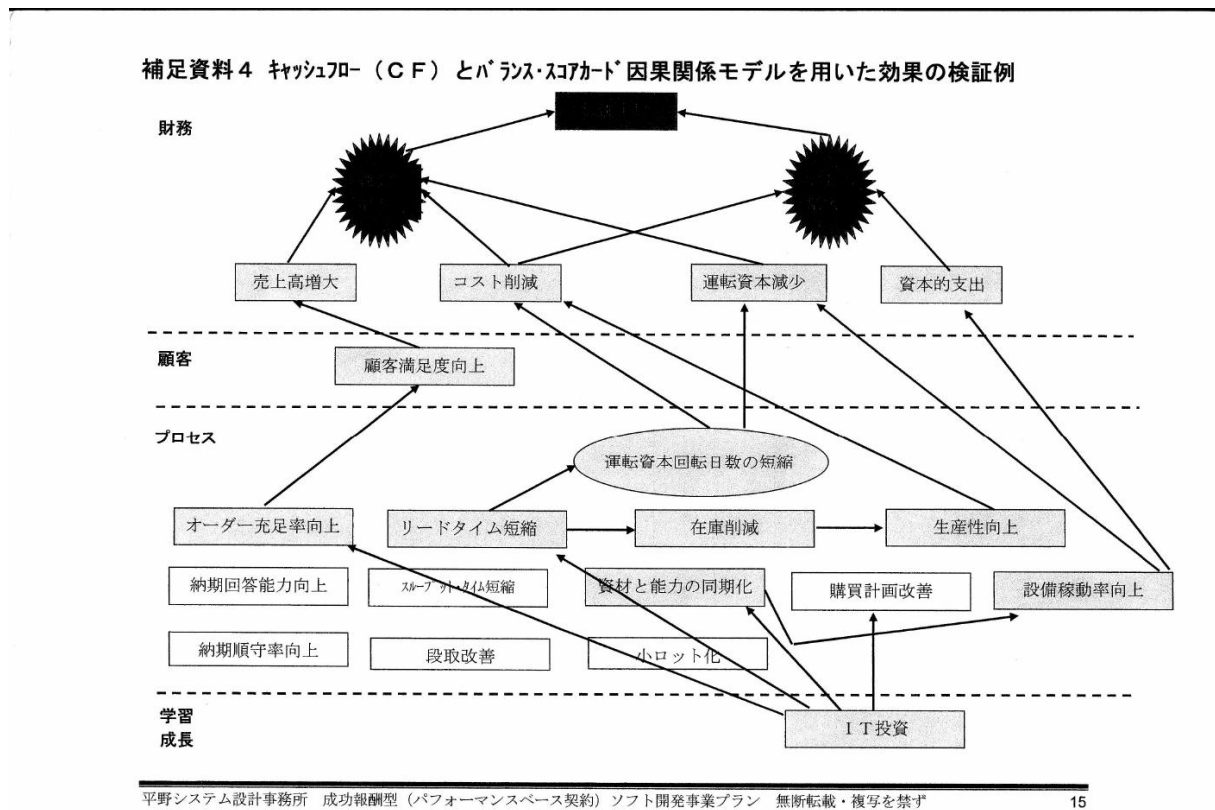
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Original diagram retained as a visual reference.

Supplemental Effect-Verification Example — Cash Flow and Causal Relationships

The final diagram traces relationships among cost reduction, reduced working capital, capital expenditure, customer satisfaction, inventory reduction, productivity, delivery response, equipment utilization, learning and other factors, linking operational effects to financial outcomes.



Original causal-model diagram retained as a visual reference.